

## YEAR 7 • UNIT 1 • HOMEWORK 1

# Integers and Cells

Mathematics 1.1 — Adding &amp; Subtracting Integers

Science 1.1 — Cells and Microscopes

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date given: \_\_\_\_\_

Date due: \_\_\_\_\_

**How to do this homework**

- Write your answers **on this sheet**, in the spaces given. If you need more room, use the back of the page.
- In maths, **write the calculation before the answer**.
- In science, answer in **full English sentences**. “Chloroplast” is not a sentence; “The green circles are chloroplasts” is.
- If a word is new to you, look for the blue **Word help** box on that page before you give up.

**Section A • Mathematics 1.1 — Adding and Subtracting Integers****Remember from the lesson**

An **integer** is a whole number that is positive, negative or zero — never a fraction. On the number line, **negative** integers sit to the **left** of zero and **positive** integers to the **right**.

Add a **positive**      move **right**

Add a **negative**      move **left**

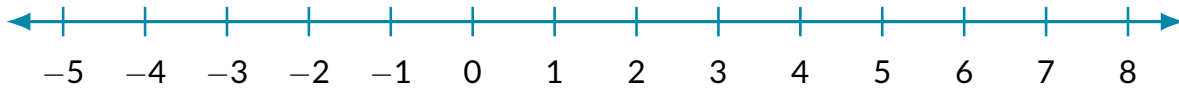
Subtract a **positive**      move **left**

Subtract a **negative**      move **right** — because  $a - (-b) = a + b$

Careful: only  $-(-)$  becomes  $+$ . A single  $+(-)$  still sends you left, so  $5 + (-3) = 2$ , **not** 8.

**A1 Show it on the number line**

(a) Show  $4 + (-3)$  on the number line below. Put a cross on the number you **start** at, then draw an arrow for the move.



$$4 + (-3) = \underline{\hspace{2cm}}$$

(b) Complete each sentence with **higher than**, **lower than**, **warmer than** or **colder than**.

(i)  $-8$  is \_\_\_\_\_  $-4$ .

(ii) A temperature of  $3^{\circ}\text{C}$  is \_\_\_\_\_ a temperature of  $-9^{\circ}\text{C}$ .

(iii) Floor  $-2$  of the car park is \_\_\_\_\_ floor 1.

**A2 Moving along the line**

Work these out. You may draw a small number line beside any of them.

(a)  $-6 + 4 = \underline{\hspace{2cm}}$

(e)  $8 + (-8) = \underline{\hspace{2cm}}$

(i)  $4 - 11 = \underline{\hspace{2cm}}$

(b)  $5 + (-9) = \underline{\hspace{2cm}}$

(f)  $-10 - (-4) = \underline{\hspace{2cm}}$

(j)  $-5 - (-5) = \underline{\hspace{2cm}}$

(c)  $-3 - 5 = \underline{\hspace{2cm}}$

(g)  $0 - (-6) = \underline{\hspace{2cm}}$

(k)  $6 - 8 - (-5) = \underline{\hspace{2cm}}$

(d)  $-2 - (-7) = \underline{\hspace{2cm}}$

(h)  $-7 + 15 = \underline{\hspace{2cm}}$

(l)  $-9 - 6 + (-1) = \underline{\hspace{2cm}}$

**A3 Say it, then write it**
**Word help – these words decide the order of your calculation**

|                            |                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>subtract 5 from 8</b>   | you <b>start</b> at 8: $8 - 5$ . The English gives the numbers in the <b>opposite order</b> to the calculation. |
| <b>take 6 away from 2</b>  | the same trap: $2 - 6$ .                                                                                        |
| <b>7 less than 3</b>       | again the same trap: $3 - 7$ .                                                                                  |
| <b>rises by / falls by</b> | the temperature goes up / the temperature goes down.                                                            |
| <b>deposit</b>             | to put money <b>into</b> an account (+).                                                                        |
| <b>withdraw</b>            | to take money <b>out of</b> an account (-).                                                                     |
| <b>owe / a debt</b>        | money you must pay back. It makes your total <b>negative</b> .                                                  |
| <b>descend</b>             | to go <b>down</b> . To go up is to <b>ascend</b> .                                                              |

Turn each English sentence into a calculation, then work out the answer. **Write the calculation first.**

1. Subtract 9 from 4.

Calculation: \_\_\_\_\_ Answer: \_\_\_\_\_

2. Take 6 away from  $-2$ .

Calculation: \_\_\_\_\_ Answer: \_\_\_\_\_

3. What is 7 less than 3?

Calculation: \_\_\_\_\_ Answer: \_\_\_\_\_

4. The temperature is  $-8^{\circ}\text{C}$  and it rises by 5 degrees.

Calculation: \_\_\_\_\_ Answer: \_\_\_\_\_

5. A submarine is at  $-30$  m. It descends a further 12 m.

Calculation: \_\_\_\_\_ Answer: \_\_\_\_\_

6. An account holds  $-35$  dollars. The bank removes a debt of 20 dollars from it.

Calculation: \_\_\_\_\_ Answer: \_\_\_\_\_

## A4 Difference

**Remember: a difference is a gap**

The **difference** between two numbers is how far apart they are on the line. Work it out with **bigger – smaller**. A difference is **never negative**. **How much warmer, how much colder, how much higher** and **how many more** all ask for the same thing.

1. Find the difference between  $-6$  and 2.

Answer: \_\_\_\_\_

2. Find the difference between  $-9$  and  $-3$ .

Answer: \_\_\_\_\_

3. How much warmer is  $4^{\circ}\text{C}$  than  $-13^{\circ}\text{C}$ ? Show your calculation.

\_\_\_\_\_

4. These two questions look almost the same. They are not.

(a) Find the difference between  $-5$  and 6.

Answer: \_\_\_\_\_

(b) What is  $-5$  minus 6?

Answer: \_\_\_\_\_

(c) In one full sentence, explain why the two answers are different.

\_\_\_\_\_

\_\_\_\_\_

5. One winter morning it is  $18^{\circ}\text{C}$  in Ho Chi Minh City and  $-24^{\circ}\text{C}$  in Ulaanbaatar. How much colder is Ulaanbaatar? Show your calculation.

## A5 Word problems

Read each one **all the way to the end** before you start writing. Show the calculation, then the answer.

1. **The office lift.** Mr Bowen gets into the lift on floor 7. He goes **down** 11 floors, and then **up** 2 floors. Which floor is he on now?

2. **A day in Sa Pa.** At 6 a.m. the temperature is  $-3^{\circ}\text{C}$ . By noon it has risen 11 degrees. By midnight it has fallen 14 degrees.

(a) What is the temperature at midnight?

(b) What is the **difference** between the warmest and the coldest temperature of the day?

3. **Six weeks of saving.** Mr Bowen's account is at  $-40$  dollars. Every Monday he deposits 25 dollars. Every Friday he withdraws 25 dollars. How much is in the account after 6 weeks? Explain your answer in one full sentence.

## A6 Now you write the question

### Your turn to be the teacher

So far the questions have given you the English, and you have written the maths. Now do it **backwards**. You are given the maths — write the English.

Your word problem must:

- be about something **real** — money, temperature, floors in a building, or depth under the sea;

- use at least one word from the **Word help** box on the earlier page, such as **rises by**, **falls by**, **deposits**, **withdraws**, **owes**, **below** or **descends**;
- be written in **full English sentences**, and end with a question;
- have the answer that is already given to you.

(a) Write a word problem for

$$-5 + 8 = 3$$

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(b) Write a word problem for

$$12 - (-4) = 16$$

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## Section B · Science 1.1 – Cells and Microscopes

### Remember from the lesson

A **cell** is the smallest basic unit of all living organisms. An **organelle** is a tiny structure inside a cell that does one specific, important job to keep the cell alive. Every cell has a **cell membrane**, **cytoplasm**, a **nucleus** and **mitochondria**. A plant cell has all of those **plus** a **cell wall** made of **cellulose**, **chloroplasts** containing **chlorophyll**, and a **sap vacuole**.

**B1 Match the word to its job**

Write the correct letter in the box next to each word.

| Word            | Letter               | Job                                                                      |
|-----------------|----------------------|--------------------------------------------------------------------------|
| 1. Cell         | <input type="text"/> | <b>A</b> The control centre of the cell – the “boss”.                    |
| 2. Organelle    | <input type="text"/> | <b>B</b> A large space of sugars and water that keeps a plant cell firm. |
| 3. Nucleus      | <input type="text"/> | <b>C</b> Where energy is released from food.                             |
| 4. Mitochondria | <input type="text"/> | <b>D</b> The smallest basic unit of all living organisms.                |
| 5. Cell wall    | <input type="text"/> | <b>E</b> A tool that uses lenses to bend light and magnify an image.     |
| 6. Chloroplast  | <input type="text"/> | <b>F</b> A tiny structure inside a cell that does one specific job.      |
| 7. Sap vacuole  | <input type="text"/> | <b>G</b> A strong, stiff outer layer that holds a plant cell in shape.   |
| 8. Microscope   | <input type="text"/> | <b>H</b> Where a plant makes its food, using sunlight.                   |

**B2 Animal, plant, or both?**

Put a tick (✓) in **one** column for each organelle.

| Organelle     | Animal cells only | Plant cells only | Both |
|---------------|-------------------|------------------|------|
| Cell membrane |                   |                  |      |
| Cytoplasm     |                   |                  |      |
| Nucleus       |                   |                  |      |
| Mitochondria  |                   |                  |      |
| Cell wall     |                   |                  |      |
| Chloroplast   |                   |                  |      |
| Sap vacuole   |                   |                  |      |

**B3 Answer in full sentences**

**Sentence starters – you may copy these**

“A plant cell needs a cell wall because...”

“One difference is that... Another difference is that...”

“A microscope is a tool that...”

1. An animal has a skeleton to hold it up. A plant does not. Explain why every plant cell needs a cell wall.

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2. Give **two** ways a cell wall is different from a cell membrane. Write each one as a full sentence.

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3. Under the microscope, a leaf cell is full of small green circles. Name them, and name the green substance inside them.

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4. What does a microscope do? Use the words **lens** and **magnify** in your answer.

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#### **B4** A model of a cell

Mr Bowen builds a model of an animal cell in his kitchen. He uses a clear plastic bag, filled with jelly, with a plum and four red beans pushed into the middle of it.

- (a) Which organelle does each item represent?

| Item in the model | Organelle it represents |
|-------------------|-------------------------|
| The plastic bag   |                         |
| The jelly         |                         |
| The plum          |                         |
| The four beans    |                         |

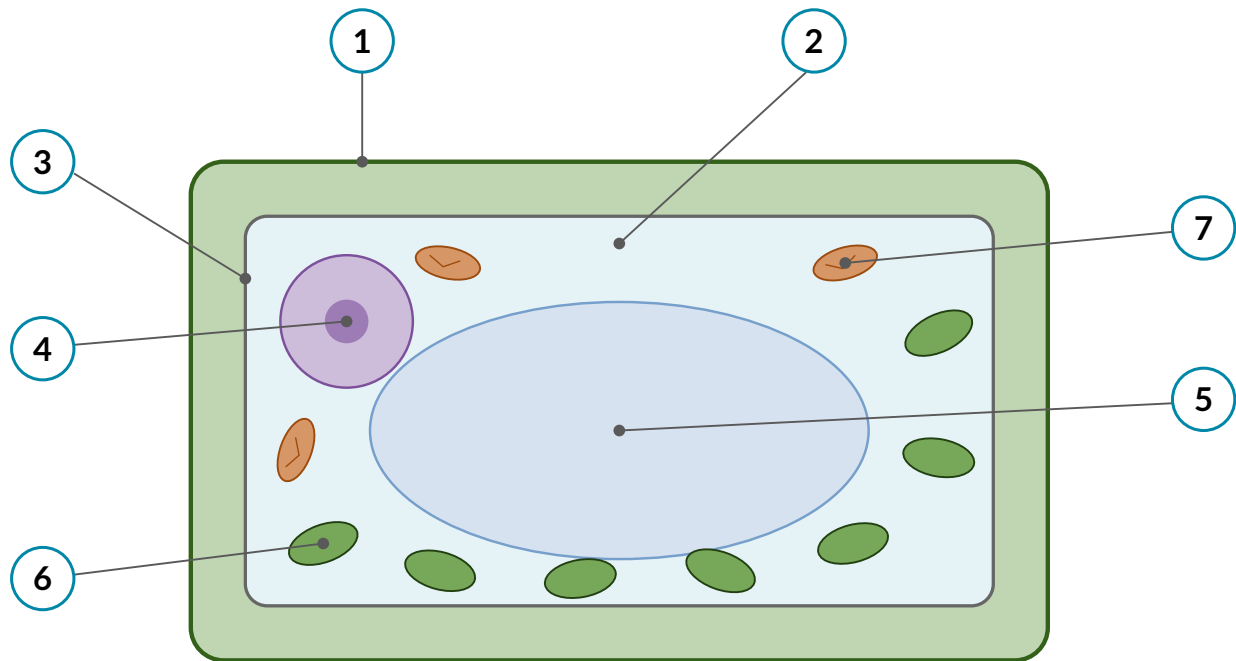
- (b) Name **one** thing he could add to turn it into a model of a **plant** cell, and say which organelle it would represent.

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**B5 Label the plant cell**

Write the name of each numbered part on the lines below the diagram.



1. \_\_\_\_\_

5. \_\_\_\_\_

2. \_\_\_\_\_

6. \_\_\_\_\_

3. \_\_\_\_\_

7. \_\_\_\_\_

4. \_\_\_\_\_

Hint: number 3 is the very thin layer just inside the outer green edge. Numbers 1 and 3 are easy to confuse — one is thick and stiff, one is thin and flexible.

## Answer Key – Teacher Copy

### A1 Show it on the number line

(a) Cross on 4, arrow three steps *left*, landing on 1. So  $4 + (-3) = 1$ . The cross must be on 4, not on 3 – adding a negative means starting at the first number and moving left.

(b) (i) lower than (ii) warmer than (iii) lower than. (“higher than” is also acceptable in (iii) if the sentence is reversed – mark the reasoning, not the word.)

### A2 Moving along the line

(a)  $-2$  (b)  $-4$  (c)  $-8$  (d)  $5$  (e)  $0$  (f)  $-6$

(g)  $6$  (h)  $8$  (i)  $-7$  (j)  $0$  (k)  $3$  (l)  $-16$

Common errors: (d) and (f) – students flip *any* pair of signs to  $+$ . Only  $-(-)$  becomes  $+$ . (b)  $5 + (-9)$  is a single  $+(-)$ , so it still goes left.

### A3 Say it, then write it

1.  $4 - 9 = -5$

2.  $-2 - 6 = -8$

3.  $3 - 7 = -4$

4.  $-8 + 5 = -3$

5.  $-30 - 12 = -42$

6.  $-35 - (-20) = -15$

Questions 1–3 are the “reversed order” trap; 6 is the “removing a debt adds” rule – the only place it is tested now, so it is worth checking. If a student writes  $9 - 4$ ,  $6 - (-2)$  or  $7 - 3$ , that is the English error, not an arithmetic one – say so, and make them re-read the sentence.

### A4 Difference

1.  $8$

2.  $6$

3.  $4 - (-13) = 17$  degrees

4. (a)  $11$  (b)  $-11$  (c) A difference is the gap between the numbers so it is never negative, but  $-5 - 6$  is an instruction to move 6 further left.

5.  $18 - (-24) = 42$  degrees

### A5 Word problems

1.  $7 - 11 + 2 = -2$ . Floor  $-2$  (B2).

2. (a)  $-3 + 11 - 14 = -6$  °C. (b) Warmest  $8$  °C at noon, coldest  $-6$  °C:  $8 - (-6) = 14$  degrees.

3. Still  $-40$  dollars. Each week  $+25 - 25 = 0$ , so six weeks change nothing. This one rewards reading to the end before calculating.

## A6 Now you write the question

There is no single right answer — check the four conditions in the green box, in this order:

1. **Does it start in the right place?** For (a) the story must *begin* at  $-5$ , not arrive there. This is the same “which number comes first” problem as A3, and it is the one most of them will get wrong.
2. **Does the signal word match the sign?** (a) needs an *up* word (rises, deposits, climbs) for the  $+8$ . (b) needs a *removing a negative* — a debt cancelled, a weight taken off — not simply “adds 4”.
3. **Full sentences, ending in a question?**
4. **Does their own answer come out as 3, and as 16?**

Two that work: (a) “The temperature in Sa Pa was  $-5^{\circ}\text{C}$  at midnight. By noon it had risen by 8 degrees. What was the temperature at noon?” (b) “Mr Bowen had 12 dollars, and he also owed his brother 4 dollars. His brother told him to forget the debt. How much is Mr Bowen worth now?”

Part (b) is the harder one by a distance — expect most of the class to write a plain addition. That is worth a whole-class discussion rather than a red pen: ask them what in real life makes you *richer* when it is taken away.

## B1 Match the word to its job

1-D 2-F 3-A 4-C 5-G 6-H 7-B 8-E

## B2 Animal, plant, or both?

Both: cell membrane, cytoplasm, nucleus, mitochondria.

Plant only: cell wall, chloroplast, sap vacuole.

Nothing is animal-only. Expect some students to tick that column anyway — it is worth saying out loud that a plant cell has everything an animal cell has, *plus* extras.

## B3 Answer in full sentences

1. A plant has no skeleton, so each cell needs a stiff wall around it to hold its shape and hold the plant up.
2. Any two: the wall is thick and stiff while the membrane is thin and flexible; the wall is made of cellulose; the wall is only in plant cells while every cell has a membrane; the membrane controls what enters and leaves, the wall does not.
3. They are chloroplasts, and the green substance inside them is chlorophyll.
4. A microscope is a tool that uses curved pieces of glass called lenses to bend light and magnify an image, making it look much bigger than it really is.

## B4 A model of a cell

(a) bag — cell membrane; jelly — cytoplasm; plum — nucleus; beans — mitochondria.

(b) Any one, with its organelle named: a shoebox or rigid container around the bag — cell wall; a water balloon in the middle — sap vacuole; green sweets — chloroplasts.

## B5 Label the plant cell

1 — cell wall 2 — cytoplasm 3 — cell membrane 4 — nucleus 5 — sap vacuole 6 — chloroplast 7 — mitochondrion